



2.4 Cubic and other polynomial models

Functions · IB Diploma · Concept guide and original exam-style practice

Understand the concept

A cubic model $y=ax^3+bx^2+cx+d$ can have up to two turning points and a change in concavity, making it suitable for data whose rate of change itself increases then decreases (or vice versa) — something a linear or quadratic model cannot capture. End behaviour is determined by the sign of the leading coefficient a and the degree: for a cubic with $a>0$, $y\rightarrow+\infty$ as $x\rightarrow+\infty$ and $y\rightarrow-\infty$ as $x\rightarrow-\infty$ (the reverse for $a<0$). Roots and turning points are typically found using technology (GDC).

Worked example: follow the reasoning

An open box is formed from a 20 cm square sheet by cutting squares of side x cm from each corner and folding up the sides, giving volume $V(x)=x(20-2x)^2$ cm³ for $0<x<10$. Using technology, find the value of x that maximizes the volume, and the maximum volume, to 1 decimal place.

Solution: Expanding, $V(x)=4x^3-80x^2+400x$. Using a GDC's maximum-finder (or solving $V'(x)=12x^2-160x+400=0 \rightarrow 3x^2-40x+100=0 \rightarrow x=(40\pm20)/6$), the valid root in $(0,10)$ is $x=10/3\approx3.33$ cm, giving $V=(10/3)(20-20/3)^2=16000/27\approx592.6$ cm³.

Quick check · multiple choice

A cubic function $y=ax^3+bx^2+cx+d$ has $a<0$. Describe its end behaviour.

- A Both ends go to $+\infty$
- B Both ends go to $-\infty$
- C As $x\rightarrow\infty$, $y\rightarrow-\infty$ and as $x\rightarrow-\infty$, $y\rightarrow+\infty$
- D As $x\rightarrow\infty$, $y\rightarrow+\infty$ and as $x\rightarrow-\infty$, $y\rightarrow-\infty$

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 A manufacturer models the cost, in thousands of dollars, of producing x hundred units of a component by $C(x)=2x^3-21x^2+60x+100$, for $0\leq x\leq 6$. (a) Find $C(2)$ and interpret this value in context. (b) Using technology, find the local minimum value of $C(x)$ for x between 3 and 6, and the corresponding value of x , to 1 decimal place. (c) State one limitation of using this cubic model for $x>6$. [7]

Final answer: _____



Answers and explanation

Question 1 [7]

(a) $C(2)=16-84+120+100=152$, meaning producing 200 units costs \$152,000. (b) $C'(x)=6x^2-42x+60=0 \rightarrow x^2-7x+10=0 \rightarrow (x-2)(x-5)=0$, so $x=2$ (local max, since $C''(2)=-18<0$) and $x=5$ (local min, since $C''(5)=18>0$). $C(5)=250-525+300+100=125$, so the local minimum cost is \$125,000 at $x=5$ (500 units). (c) The cubic model was fitted only for $0 \leq x \leq 6$; extrapolating beyond $x=6$ may not reflect real production costs, since the underlying cost behaviour (e.g. economies/diseconomies of scale) may change outside this range.

Quick check answer

C. With a negative leading coefficient, the cubic rises on the left and falls on the right.

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Mathematics AI subject guide.

Original FagusED teaching material. Not an official IB publication; question style informed by IB specimen papers but content and numbers are original.